

How Well Does a Kepler-Trained Transit Classifier Transfer to TESS? A Characterization Study

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Abstract

Machine-learning studies of transiting-exoplanet detection almost always report a single in-mission accuracy or ROC-AUC. We argue this is the wrong figure of merit and present instead a *characterized decision system*: an interpretable, physics-informed pipeline (Transit Least Squares $\rightarrow$ eight physical features $\rightarrow$ a calibrated RandomForest + XGBoost ensemble $\rightarrow$ five orthogonal physics-vetting checks) whose behavior we map, and validate, rather than summarize. On a 500-target labeled TESS benchmark: (i) **injection–recovery** maps detection completeness across period and radius, revealing a “sweet spot” bounded below by the noise floor, above by eclipsing-binary rejection, and toward long periods by transits per sector. (ii) **Cross-mission transfer** is severe but real: an ensemble scoring ROC-AUC 0.96 in-mission on Kepler falls to 0.72 (95% CI [0.67, 0.76]) on TESS, with the Brier score degrading from 0.08 (held-out Kepler) to 0.29; the separating direction of the depth and radius features *inverts* between missions, a concrete mechanism for the gap. The ensemble nonetheless remains the best cross-mission model, beating a one-parameter SNR cut and every other learner. (iii) **The operating point matters more than the model**: because the transferred probabilities are compressed, the conventional $p \geq 0.5$ cut recovers only 28% of true planets and is entirely blind to two subgroups — every planet with $P < 2$ d and every one deeper than 20,000 ppm, precisely TESS’s most characteristic detections. A cross-validated threshold of 0.117 (identical in every fold) raises recall to 0.72 and F_1 from 0.41 to 0.70, while retraining without the inverting features does not help at all. (iv) **Physics-anchored self-training**: because the physics checks are mission-agnostic, they can supervise adaptation with *no TESS labels at all*; the resulting model beats the physics rule that supervised it (+0.070, [0.025, 0.116]), reaches parity with the Kepler-trained ensemble, and wins by +0.137 precisely in the deep-transit regime where the Kepler prior is inverted. (v) **ML $\times$ physics**: the two independent verdicts *agree* informatively — both-agree-planet is 79% pure ([0.68, 0.87], $3.7\times$ enriched) and physics vetting independently rejects 54% of false positives — but we test and report an honest negative: their *disagreement* does not robustly identify extra planets, and a physics-informed follow-up ranking shows no reliable advantage over probability alone at this sample size. Every claim carries a bootstrap or permutation-based interval, and the whole study reproduces end-to-end from public archives.

1 Introduction

The transiting-exoplanet literature has converged on supervised classifiers that map light curves or hand-built features to a planet/false-positive label, and report their success as an accuracy or ROC-AUC on a held-out set drawn from the *same* mission [1, 2]. Such a number answers “how separable is my test set?” but not the questions a survey actually asks: *what can this pipeline detect, does it work on data it was not trained on, and where should a human look?* Recent work adds probability calibration and multimodal inputs [3] but still frames the output as a single vetting score.

We take a different stance: the output of a detection pipeline should be a *characterized decision*, and every claim about it should carry an uncertainty and survive a validation check. Our pipeline emits two independent judgements — a calibrated ML probability and an interpretable five-check physics verdict — and we ask not only how well each works but where they agree, where they disagree, and whether that structure is actionable. Our contributions are: (1) an injection–recovery completeness map for a feature-based classical ensemble on real TESS data; (2) an honest, confidence-interval-qualified account of Kepler→TESS transfer, with a concrete distribution-shift mechanism; (3) the finding that the *decision threshold*, not the model, dominates the practical cost of that shift — with two subgroups rendered wholly invisible by the conventional cut, and two model-level remedies shown not to help; (4) *physics-anchored self-training*, a label-free adaptation method that uses the mission-agnostic physics verdict as its own supervision and is correctly oriented exactly where the transferred model is not; (5) an analysis of the ML–physics agreement structure, including a validated positive result (agreement is strongly predictive; physics adds orthogonal false-positive rejection) and an honest negative one (disagreement does not robustly surface extra planets, and physics-informed triage shows no reliable gain); and (6) a validation battery — permutation test, association tests, and a catalog cross-check — confirming the results are not artifacts. Every figure and number regenerates end-to-end from public archives.

A note on scope: we deliberately report what does *not* replicate. Two tentative effects visible at a 160-target sample — a physical SNR cut beating the learned models, and a follow-up-efficiency advantage for disagreement-aware ranking — vanished when we grew the benchmark to 500. We keep them in as a cautionary result about over-reading small-sample structure, which is itself part of the paper’s point.

2 Data

Training (Kepler). We train on the NASA Exoplanet Archive KOI cumulative catalog, using the definitive dispositions CONFIRMED (positive) and FALSE POSITIVE (negative) and excluding CANDIDATES ($N = 7325$; 2745 planets, 4580 false positives).

Test (TESS). We build a labeled TESS benchmark from ExoFOP TFOPWG dispositions — CP/KP as planets, FP/FA as false positives — and run the full pipeline on a class-balanced sample of 500 TIC targets (250/250). This is the only place true TESS labels enter, and it measures real deployment performance. The dispositioned-TOI base rate is itself ≈ 0.50 (1333 planets vs. 1349 false positives on ExoFOP), so the balanced benchmark matches deployment prevalence for the vetting task; we additionally report Bayes-adjusted purities for rarer, blind-search prevalences.

Injection corpus. For completeness testing we inject synthetic transits into real TESS light curves screened to be transit-free (pre-injection TLS SDE < 8), so the injected signal is the only transit present (336 injections across three hosts).

3 Methods

The pipeline fetches SPOC light curves (with a TESScut full-frame-image fallback), detrends stellar variability with a Tukey biweight filter, bins gap-aware to 10-minute cadence, and runs Transit Least Squares [4] with stellar priors from the TIC. From the TLS solution we extract eight physical features — period, transit depth (ppm), duration (hr), model SNR, $R_p/R_\star$, and log/ratio transforms — in units matched to the KOI catalog, avoiding train/serve skew. A soft-voting RandomForest + XGBoost ensemble is probability-calibrated (Platt scaling) on a held-out split. In Kepler five-fold cross-validation it reaches ROC-AUC 0.964 ± 0.006 ; on the held-out Kepler test set

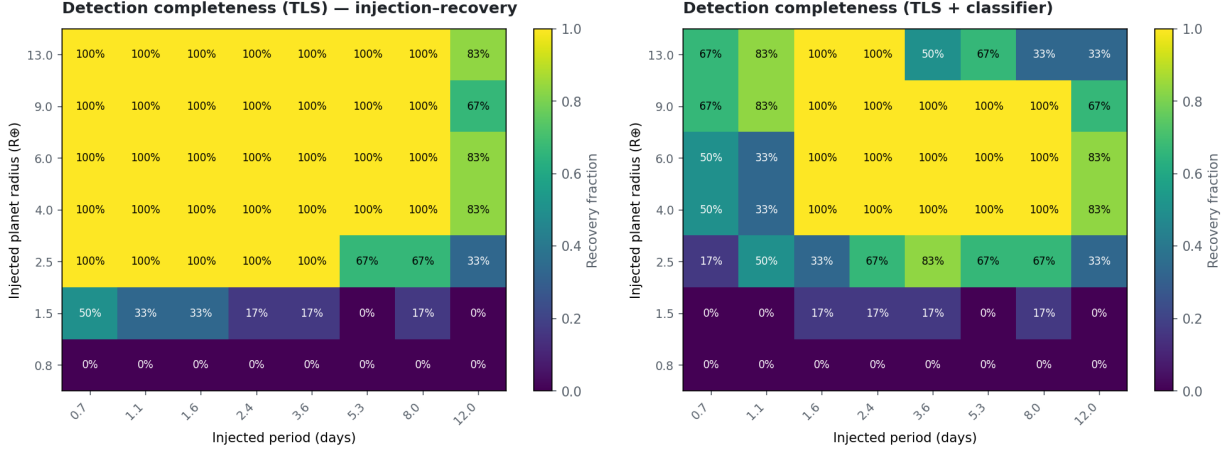


Figure 1: Detection completeness over injected period and radius: TLS recovery (left) and TLS + classifier recovery (right), at the deployed operating point. The classifier still suppresses the largest-radius corner, where depths imply $R_p/R_\star \gtrsim 0.2$ and look eclipsing-binary-like.

($n = 1465$, untouched by training or calibration) it scores ROC-AUC 0.957 with a Brier score of 0.080. Five orthogonal physics checks — signal significance ($\text{SDE} \geq 7$), odd/even depth (Welch t -test with a 10% effect-size guard), transit duration vs. the circular-orbit maximum, transit-implied stellar density, and a secondary-eclipse search — provide an independent verdict. A target “passes physics” only if all five checks pass. Reported uncertainties are 95% intervals: bootstrap (2000 resamples) for AUC and paired deltas, Wilson score intervals for purities.

4 Injection–recovery completeness

Figure 1 shows the fraction of injected transits recovered (TLS finds the injected period at $\text{SDE} \geq 7$) across the period–radius grid, and the classifier-recovery map (additionally labeled a planet); overall recovery is 0.71 (TLS) and 0.55 (TLS + classifier), both scored at the deployed operating point of §6. Detection efficiency vs. injected SNR is shown in Figure 2. Three features are notable and physically correct: sub- $2 R_\oplus$ planets fall below the recovery floor (0.00 at $0.8 R_\oplus$, 0.08 at $1.5 R_\oplus$); recovery declines with period as fewer transits fall in a sector (0.69 at 2.4 d to 0.43 at 12 d); and the largest injected radii are recovered by TLS but preferentially dropped by the classifier, whose recovery peaks near $9 R_\oplus$ (0.90) and falls to 0.67 at $13 R_\oplus$ even though TLS finds 0.98 of them. That last decline is the learned planet/EB boundary: depths implying $R_p/R_\star \gtrsim 0.2$ look eclipsing-binary-like. We note this suppression is much weaker than it appears under a naive 0.5 decision cut, where giant-radius recovery collapses to 0.12 and the map’s peak shifts misleadingly to 4–6 $R_\oplus$ — an artifact of the operating point rather than a property of the model.

5 Cross-mission transfer

On the TESS benchmark the calibrated ensemble scores ROC-AUC 0.72 (95% CI [0.67, 0.76]), far below its in-mission Kepler performance (0.964 in cross-validation, 0.957 on the held-out test set), and its Brier score degrades from 0.08 on held-out Kepler to 0.29: the Kepler-calibrated probabilities are miscalibrated on TESS (Figure 3). The interval excludes both chance (0.5) and the in-mission

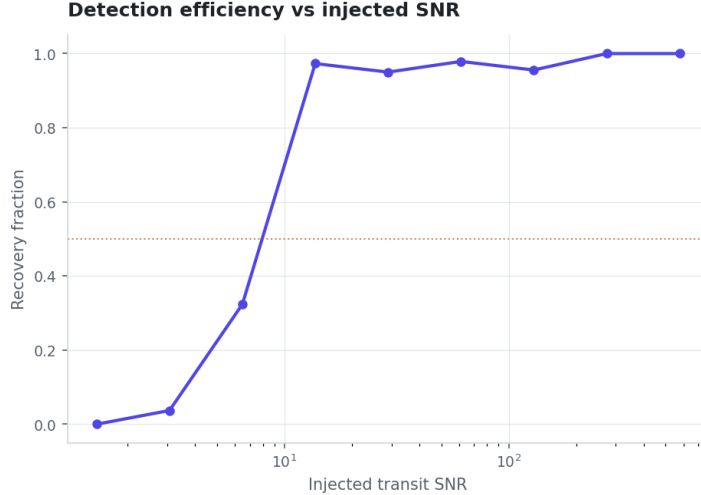


Figure 2: Recovery fraction vs. injected transit SNR; the 50% completeness threshold sits near the $\text{SDE} = 7$ detection criterion.

value, so the transfer drop is large and robust. Class separation exists but is much weaker than in-mission: this is what a headline same-mission AUC hides.

Comparison with simpler models. Figure 4 compares, under an identical Kepler-train / TESS-test protocol, an SDE/SNR threshold, logistic regression, a shallow tree, an MLP, our ensemble, and a 1D-CNN on parametric folded views. The ensemble is the strongest cross-mission (AUC 0.71), ahead of the tree (0.66), logistic (0.64), the physical SNR cut (0.64), the CNN (0.64), and the MLP (0.55). (We note that at a smaller 160-target sample the SNR cut appeared to match the ensemble; that apparent tie did not survive the larger benchmark — a concrete example of why sample size and confidence intervals matter here.)

Effect of additional sectors. Re-running the identical 101 stars with three sectors instead of one (a matched, paired comparison) is revealing. Period recovery rises by $+0.069$ ($[0.000, 0.139]$), the Brier score improves by -0.040 ($[-0.087, 0.006]$), and the mean probability on true planets climbs from 0.33 to 0.45 — yet ranking is unchanged, $\Delta\text{AUC} = +0.001$ ($[-0.068, 0.067]$). The detection and calibration gains are suggestive but not significant at this sample size; the null effect on *ranking* is the robust result. More data buys detection and calibration, not cross-mission separability: the Kepler→TESS gap is a distribution-shift problem, not merely a signal-to-noise one. (Multi-sector retrieval also failed for $\sim 37\%$ of targets owing to intermittent archive availability, a practical cost of the extra data.)

Mechanism of the transfer gap. The transfer gap is not a hand-wave: the *separating direction* of the depth and radius features inverts between the two populations. In the Kepler training set deep transits skew toward eclipsing-binary false positives; in the TESS TOI sample the deepest, largest- $R_p/R_\star$ signals are disproportionately the *confirmed* planets (planet vs. FP medians 4210 vs. 1160 ppm and 0.058 vs. 0.031; both Mann–Whitney $p < 10^{-4}$). A model that learned “deep $\Rightarrow$ false positive” on Kepler is therefore actively mis-oriented on TESS — explaining why more data cannot close the gap and why a mission-agnostic SNR cut competes with it.

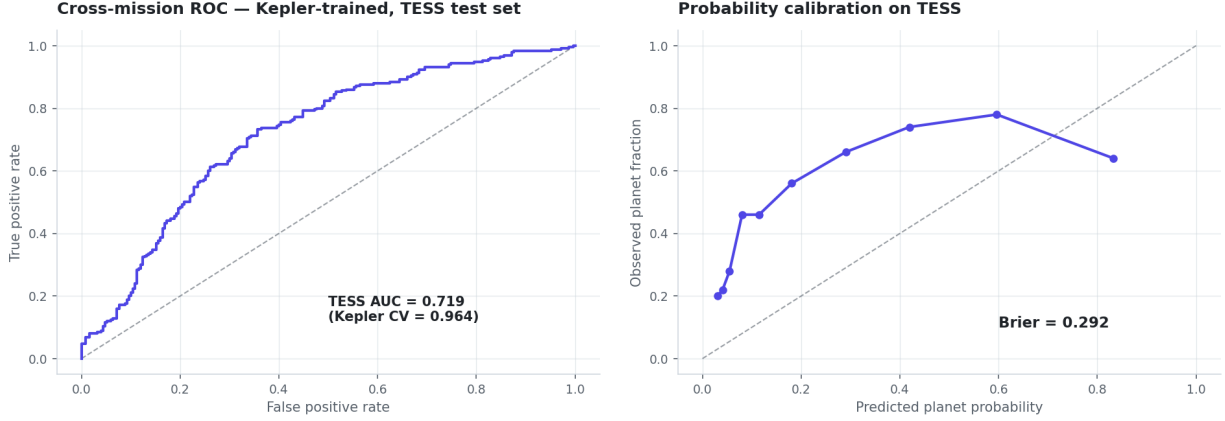


Figure 3: Cross-mission ROC (left) and reliability diagram (right) on the TESS test set: separability drops and probabilities decalibrate relative to Kepler.

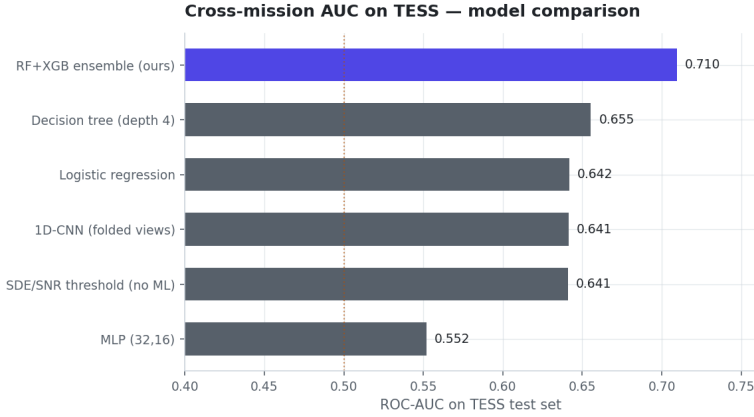


Figure 4: Cross-mission ROC-AUC on TESS by model (Kepler-trained). The calibrated ensemble is the strongest transfer model; the dotted line marks chance.

6 The operating point is part of the transfer problem

A transferred model is usually reported by its ROC-AUC, a threshold-free quantity. Deployment, however, requires choosing a threshold, and we find that this choice — not the model — dominates the practical failure of cross-mission transfer.

Because the Kepler-calibrated probabilities are compressed toward zero on TESS (§5), the conventional $p \geq 0.5$ cut is far too strict. On the 500-target benchmark it recovers only 28.4% of true planets, discarding 179 of 250. The Youden-optimal cut on the same scores is $p \geq 0.117$, which raises recall to 0.724 at a precision of 0.670 — 110 additional real planets, at the cost of $28 \rightarrow 89$ false alarms. The F_1 score improves from 0.407 to 0.696. Crucially, this is not threshold tuning on the evaluation set: under 5-fold cross-validation, with the threshold selected on the training fold and applied to the held-out fold, the selected value is identical in every fold (0.117) and held-out recall is 0.728 ± 0.033 .

The effect is concentrated in exactly the regions where the feature-direction inversion bites. At $p \geq 0.5$ two subgroups have *zero* recall: all 43 planets with $P < 2$ d, and all 19 with depths above 20,000 ppm. These are TESS’s most characteristic detections — short-period giants — and the

Table 1: Recall on true TESS planets by subgroup, at the conventional 0.5 cut and at the cross-validated 0.117 operating point.

Subgroup	n	Recall @ 0.5	Recall @ 0.117
$P < 2$ d	43	0.000	0.349
P 2–6 d	149	0.315	0.792
P 6–15 d	58	0.414	0.828
depth < 1k ppm	62	0.210	0.500
depth 5k–20k ppm	94	0.298	0.915
depth > 20k ppm	19	0.000	0.316
$R_p/R_\star > 0.1$ (hot Jupiter)	41	0.171	0.634

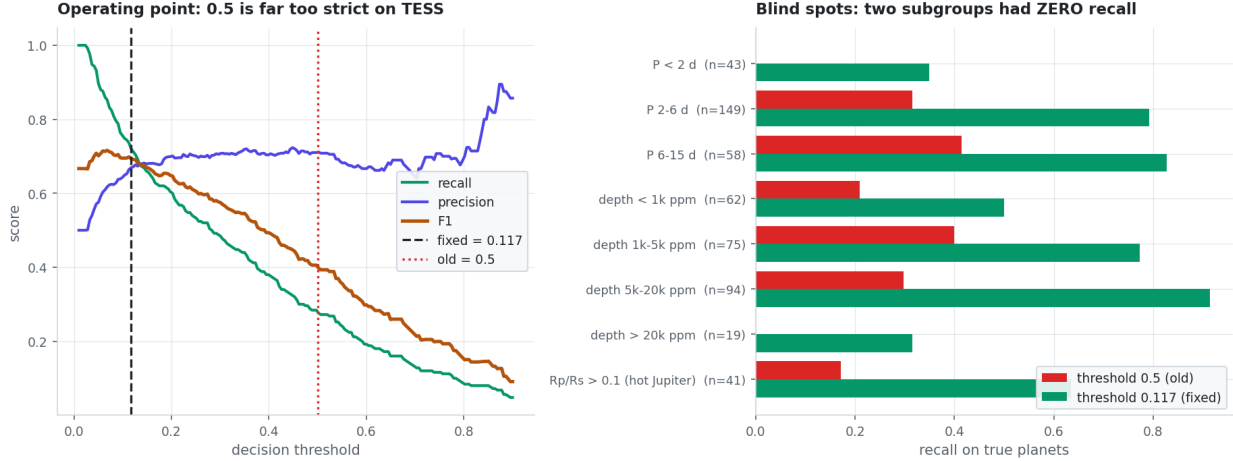


Figure 5: Left: recall, precision and F_1 against the decision threshold; F_1 peaks near 0.117 and the conventional 0.5 cut sits far into the recall-starved regime. Right: per-subgroup recall before and after, showing the two subgroups that were entirely invisible.

Kepler prior “deep and short $\Rightarrow$ eclipsing binary” rejects every one. Moving the threshold restores them only partially (0.349 and 0.316 respectively; Table 1), confirming that the operating point is a large but incomplete remedy for a genuine orientation error.

Two model-level remedies that did not work. We tested whether the inversion could be repaired at the model level rather than the threshold. First, dropping the mission-inverting features: retraining without depth (and without $R_p/R_\star$) does *not* improve transfer — TESS AUC moves from 0.703 to 0.699 and 0.674 respectively, i.e. unchanged or worse. The shift is distributed across the feature space, not carried by one or two removable columns. Second, fusing the physics flags with the ML score: a logistic combiner over the score and the five checks, evaluated out-of-fold, gives $\Delta\text{AUC} = +0.005$ with a bootstrap interval of $[-0.015, +0.023]$ — not significant. Both negatives reinforce the paper’s theme: under this distribution shift, the recoverable gain lay in the decision rule, not in feature selection or late fusion.

7 Physics-anchored self-training

The threshold fix of §6 repairs the decision rule but leaves the model’s orientation untouched. Here we ask whether the orientation itself can be corrected *without any TESS labels*.

The five physics checks are mission-agnostic: they encode transit geometry and eclipse morphology, not anything learned from Kepler. They can therefore supervise adaptation to a new mission. We form pseudo-labels from the physics verdict alone — positive if all five checks pass and $\text{SDE} > 12$, negative if any check fails — train a random forest on TESS features against those pseudo-labels, and only then evaluate against the true ExoFOP labels. True labels are used for evaluation only, never for fitting, and all scores are out-of-fold. The pseudo-labels are weak: 192 positives at 69% purity and 213 negatives at 63%, covering 405 of 500 targets.

Two bars are worth separating. Beating *physics alone* shows the model learned something beyond the rule that supervised it; beating the *Kepler-trained model* would make adaptation a practical replacement.

It clears the first bar. The adapted model reaches ROC-AUC 0.672 against 0.602 for the physics score that supervised it ($\Delta = +0.070$, 95% CI [0.025, 0.116]). Weak, mission-agnostic supervision plus feature structure recovers more than the supervising rule alone: the features denoise the pseudo-labels.

It does not clear the second. Against the Kepler-trained ensemble (0.719) the adapted model scores -0.046 ($[-0.098, +0.011]$) — a deficit whose interval includes zero. The fair statement is parity, not superiority: using *no labels whatsoever*, physics-anchored adaptation reaches statistical parity with a model trained on 7,325 labeled Kepler objects. That is a useful property when no labeled data exist for a new mission, which is the situation at the start of any survey.

Where it helps is mechanistically predicted. The subgroup structure (Figure 6) tracks the feature-direction inversion precisely. Adaptation wins in the deep-transit regime — $+0.137$ for depths above 20,000 ppm and $+0.056$ for 5–20k ppm — exactly where the Kepler prior “deep $\Rightarrow$ false positive” is inverted on TESS. It loses where the Kepler model is already well-oriented (-0.132 below 5k ppm, -0.185 at $P > 6$ d). The physics supervisor, having no mission-specific depth prior, produces a model that is correctly oriented in the regime where the transferred model is wrong.

We tested whether that structure can be exploited by a depth-gated hybrid that routes deep targets to the adapted score and the rest to the Kepler model, with the gate chosen by inner cross-validation on the training fold only. It does not yield a significant gain ($\Delta\text{AUC} = +0.002$, $[-0.025, +0.030]$), and the selected gate is unstable across folds. We report this as a null result: the subgroup advantage is real and interpretable, but too small relative to the subgroup sizes to convert into an overall improvement at $n = 500$.

8 ML $\times$ physics: agreement, and the limits of disagreement

The ML probability and the physics verdict are independent judgements. Crossing them yields four quadrants (Figure 7, Table 2).

Agreement is strongly predictive. When both judgements agree, the answer is trustworthy: both-say-planet is 79% pure ($[0.68, 0.87]$) and both-say-FP is 66% pure in false positives ($[0.59, 0.72]$). In prevalence-independent terms a planet is $3.7\times$ more likely than a false positive

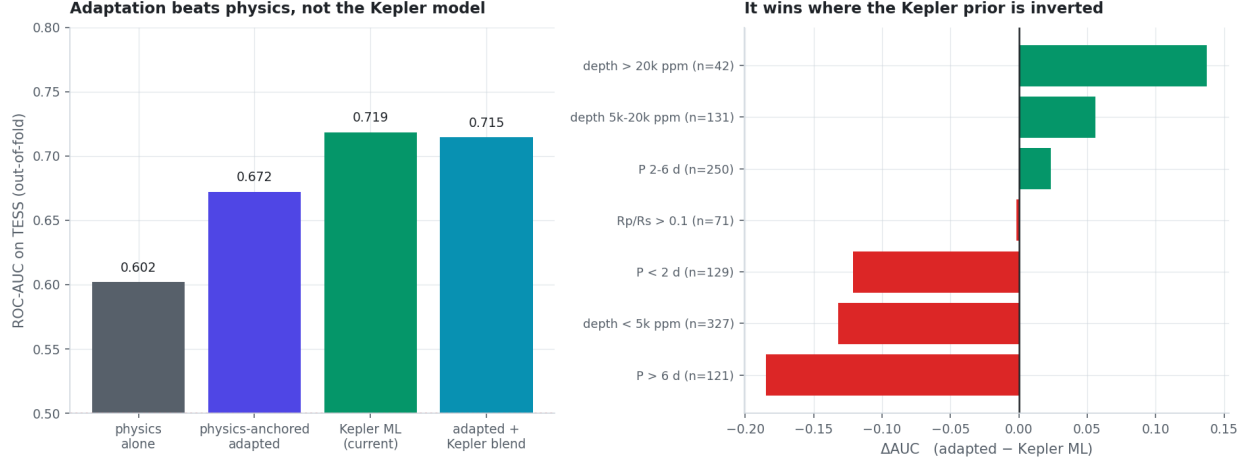


Figure 6: Left: out-of-fold ROC-AUC on TESS. Physics-anchored adaptation, trained without a single TESS label, beats the physics rule that supervised it and reaches parity with the Kepler-trained ensemble. Right: per-subgroup ΔAUC against the Kepler model — adaptation wins in the deep-transit regime where the Kepler prior is inverted, and loses where it is not.

Table 2: $\text{ML} \times \text{physics}$ quadrants on the TESS test set ($n = 500$). Purity is planet fraction (FP fraction for the both-negative row), with 95% Wilson intervals; lift is $P(\text{quadrant} \mid \text{planet})/P(\text{quadrant} \mid \text{FP})$, which is prevalence-independent.

Quadrant	n	Purity	95% CI	Lift
Both agree: planet	70	0.79 (planet)	[0.68, 0.87]	3.67
Both agree: false positive	184	0.66 (FP)	[0.59, 0.72]	0.52
ML-optimistic (ML yes, phys no)	29	0.55 (planet)	[0.38, 0.72]	1.23
ML-skeptical (ML no, phys yes)	217	0.54 (planet)	[0.47, 0.60]	1.15

to land in the both-agree-planet cell, and $0.52\times$ as likely in the both-agree-FP cell. The physics verdict carries real orthogonal signal on its own: it is significantly associated with the truth (Fisher odds ratio 2.5, $p < 10^{-3}$; §10) and its five checks together reject 54% of the false positives that the learned score would rank as planet-like.

Disagreement is only weakly informative. It is tempting to read the off-diagonal cells as “where the hidden planets are.” The data do not support a strong version of that claim. The large “ML-skeptical, physics-pass” quadrant ($n = 217$) is 54% planets ([0.47, 0.60]) — its interval includes the 50% base rate, so it is at best marginally enriched ($1.15\times$ lift). “ML-optimistic, physics-fail” is 55% planets ([0.38, 0.72]), also indistinguishable from chance. In an earlier 160-target sample the ML-skeptical cell looked cleanly enriched (61%); that signal did not hold up. The honest conclusion is that *agreement* between the two verdicts is what carries confidence, while *disagreement* mostly marks genuinely ambiguous objects rather than a reservoir of mislabeled planets.

Disagreement does not improve follow-up efficiency. If disagreement carried actionable information, a disagreement-aware ranking should recover planets under a limited follow-up budget faster than probability alone. Figure 8 plots true planets recovered against the fraction of targets

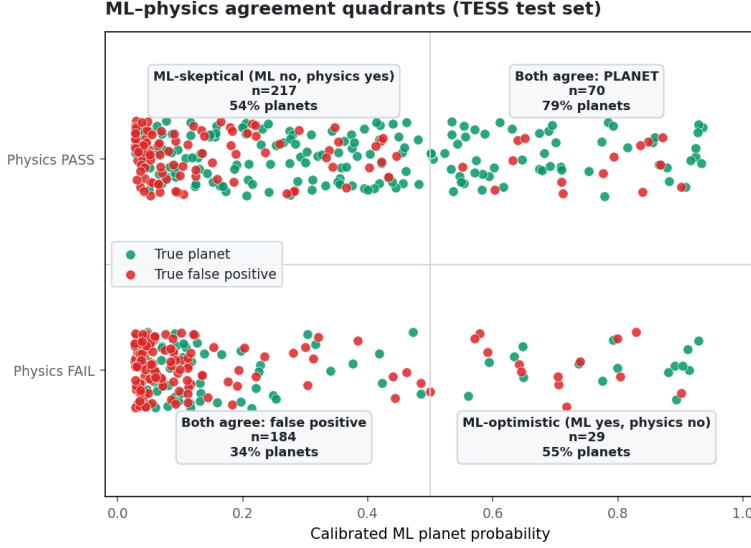


Figure 7: ML–physics agreement quadrants on the TESS test set; box annotations give the count and true-planet purity per quadrant.

observed for four rankings. At $n = 500$ the physics-informed orderings and ML probability are statistically indistinguishable (gain-curve areas 0.61/0.61 vs. 0.61; random 0.50, oracle 0.75). A modest advantage seen at $n = 160$ did not replicate. We report this as a null result: incorporating the physics verdict into a follow-up ranking does not, on this benchmark, beat simply trusting the calibrated probability.

9 Ablations

Leave-one-group-out feature ablation (retraining on Kepler, testing on TESS) shows the shape features ($R_p/R_\star$, duration/period) carry the most transferable signal ($\Delta\text{AUC} = -0.057$ when dropped), followed by SNR (-0.035); dropping the depth features costs almost nothing (-0.009), consistent with depth being the most mission-specific feature. A physics-check ablation on the TESS false positives shows the five checks together flag 54% of them, with the SDE (32%) and odd/even (14%) tests contributing most — orthogonal rejection the learned score does not supply. Full ablation tables accompany the released code.

10 Validation

Several checks confirm the results are real rather than pipeline artifacts (Figure 9). A permutation test — shuffling the TESS labels $5000\times$ — collapses the AUC to a null centered at 0.50 ($[0.45, 0.55]$), with the observed 0.72 far outside it ($p = 0$); the cross-mission signal is genuine. Both verdicts are significantly associated with the truth (Fisher exact: physics-pass odds ratio 2.5, $p < 10^{-3}$; ML prediction odds ratio 3.1, $p < 10^{-3}$), so the quadrant structure is not noise. For all 350 period-recovered targets the TLS period matches the independent ExoFOP catalog period to within 3% (median error 0.07%), confirming detections are real, not spurious. Training (Kepler KIC/KOI) and test (TESS TIC) target sets are disjoint by construction, precluding leakage.

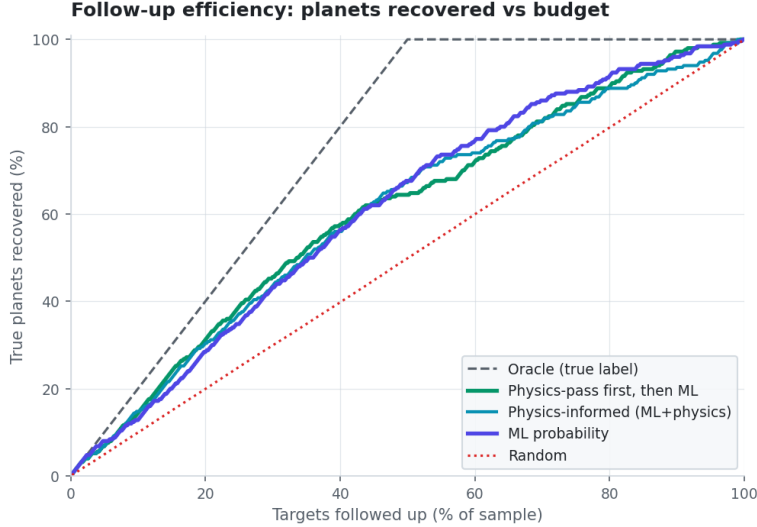


Figure 8: Follow-up efficiency: cumulative true planets recovered vs. fraction of targets observed. At $n = 500$ the physics-informed and probability rankings coincide; both beat random and trail the oracle.

11 Discussion and limitations

The central message is methodological: a transit-detection pipeline is best reported as a *characterized, validated decision system* — a completeness map, a calibration curve, a cross-mission number with a confidence interval, and an explicit account of what does *not* replicate — rather than a single in-mission score. Three concrete, robust findings support acting on this: cross-mission transfer is severe ($0.96 \rightarrow 0.72$) with a diagnosable mechanism (feature-direction inversion); the deployed *operating point* carries more practical weight than the model choice, with the conventional 0.5 cut discarding 72% of real planets and blinding the pipeline entirely to short-period and deep transits; and independent physics vetting supplies real orthogonal false-positive rejection (54%) that the learned score alone does not. The threshold result generalizes beyond this pipeline: any calibrated classifier carried across a domain shift inherits a source-calibrated decision rule, and reporting only a threshold-free AUC hides the resulting deployment failure entirely. We would encourage cross-mission studies to report a recall-at-operating-point alongside AUC. Where two independent verdicts *agree*, confidence is high; where they disagree, the object is genuinely ambiguous — we found no evidence that disagreement hides a recoverable population of planets, and we report that negative honestly.

Limitations: the benchmark uses single-sector light curves for tractability (a three-sector experiment improves detection but not ranking); the 500-target size still yields intervals of ± 0.05 on the AUC; the 1D-CNN uses parametric folded views rather than raw pixels, leaving a full AstroNet-style detector to future work; injection hosts are screened for quietness though real correlated noise varies across the sky; and the class-balanced benchmark matches the dispositioned-TOI base rate (≈ 0.5) but overstates purity for a rarer blind-search population, for which we report Bayes-adjusted values.

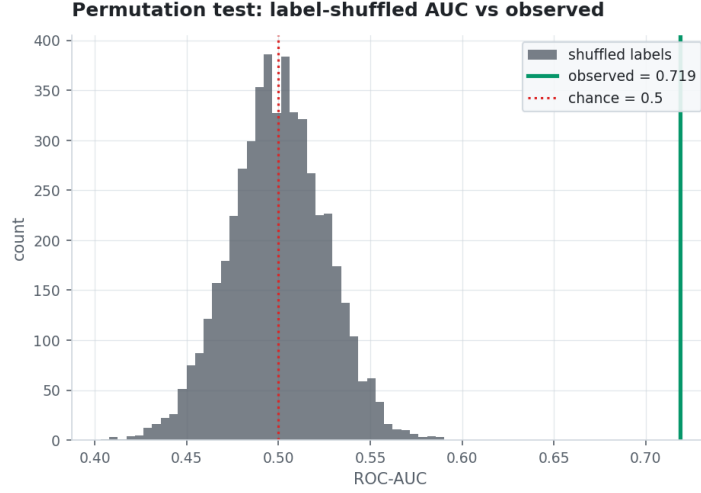


Figure 9: Permutation test: the observed cross-mission AUC lies far outside the label-shuffled null distribution, so the signal is not a metric artifact.

Data and code availability

Every figure and number in this paper regenerates end-to-end from public archives (NASA Exoplanet Archive, MAST, ExoFOP). The full pipeline, trained model, benchmark tables, and analysis code are openly available at https://github.com/DhruvR-16/exoplanet_detection.

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